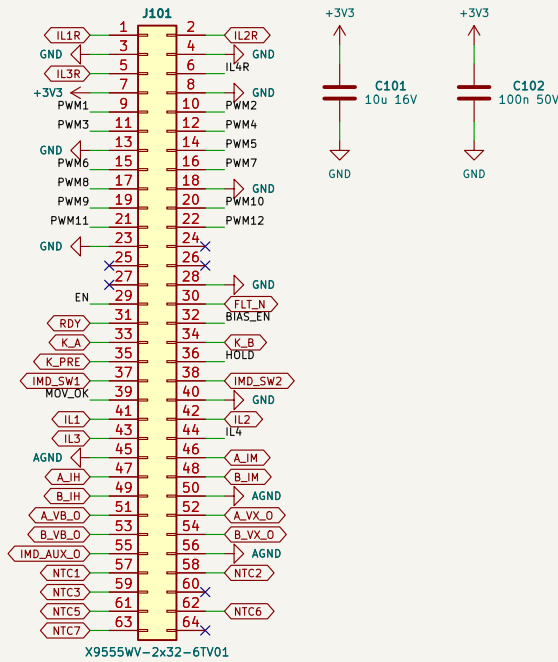
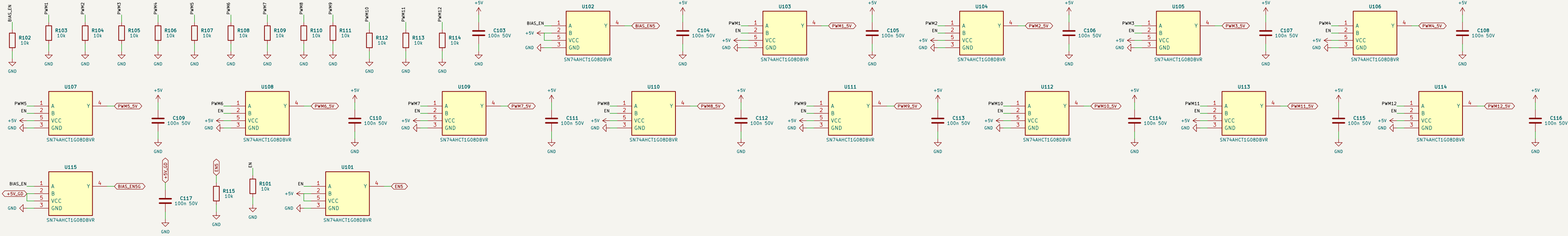


PC CONNECTOR (INTERFACES.PC), LIVE



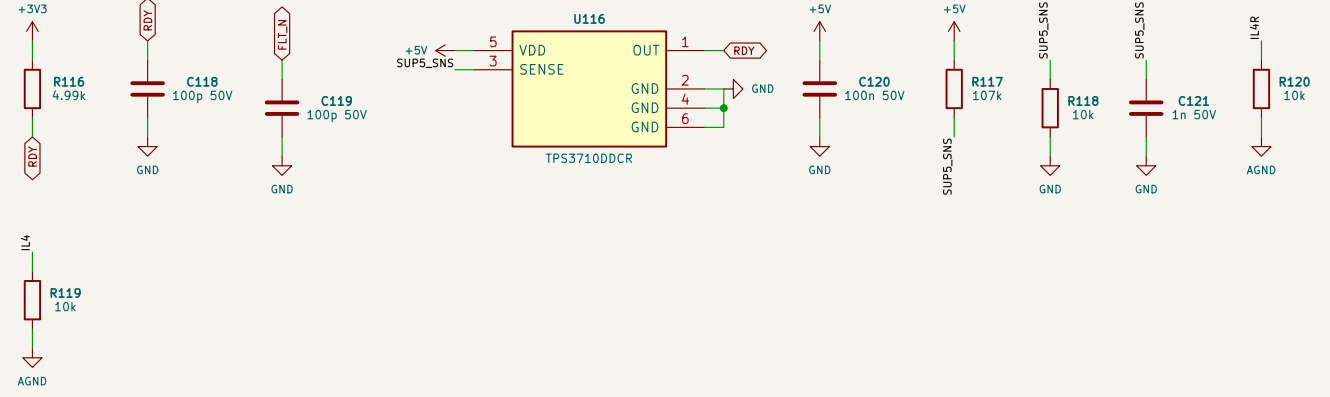
Both boards live: GND = AGND = BUS-. +3V3 goes TO the control board (rev A2; the former +24V / +5V pins carry the sensor references IL1R-IL4R). Commands are 3.3 V logic after the control board's trip latch; every command input is pulled low here (open = off).
PV-P75: PWM13-16, NTC4 and NTC6 not connected; IL4 and IL4R pulled to AGND (0 V = no phase 4).

COMMAND INPUTS: PULL-DOWNS AND 3.3 -> 5 V BUFFERS (GDRV.INPUT_BUFFERS)



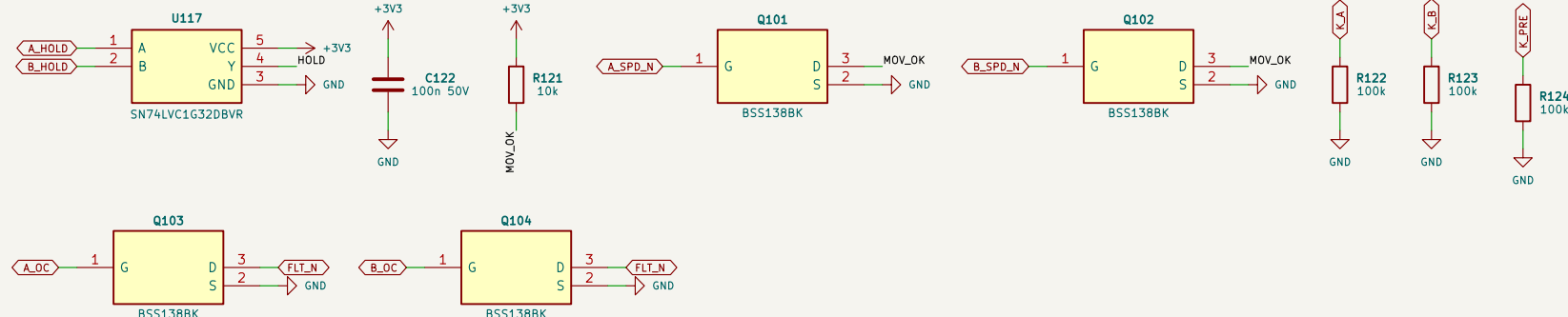
NSI6651 inputs are specified at VCC1 = 5 V only (VINH <= 3.5 V, p7) and the SN6505B EN needs 0.7 VCC: one SN74AHCT1G08 (VIN 2.0 V) per line. Each PWM is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48-0.8 us).

RDY (WIRED-AND) AND FLT_N (WIRED-OR)



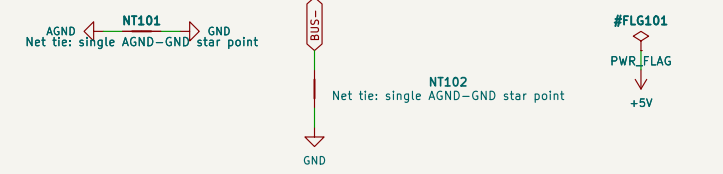
RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT + port OC ([1] >= 387-413 A, lean_shunt) pulls FLT_N low (open drain); the control board's latch trips without firmware (4R4-fuse gap, port-4prc).
+5V supervisor TPS3710 107k/10k; RDY low below 4.52-4.69 V (AHC1 VCC <= 4.5 V, bias headroom).

PORT STATUS TO PC: HOLD, MOV_OK, PORT OVER-CURRENT INTO FLT_N

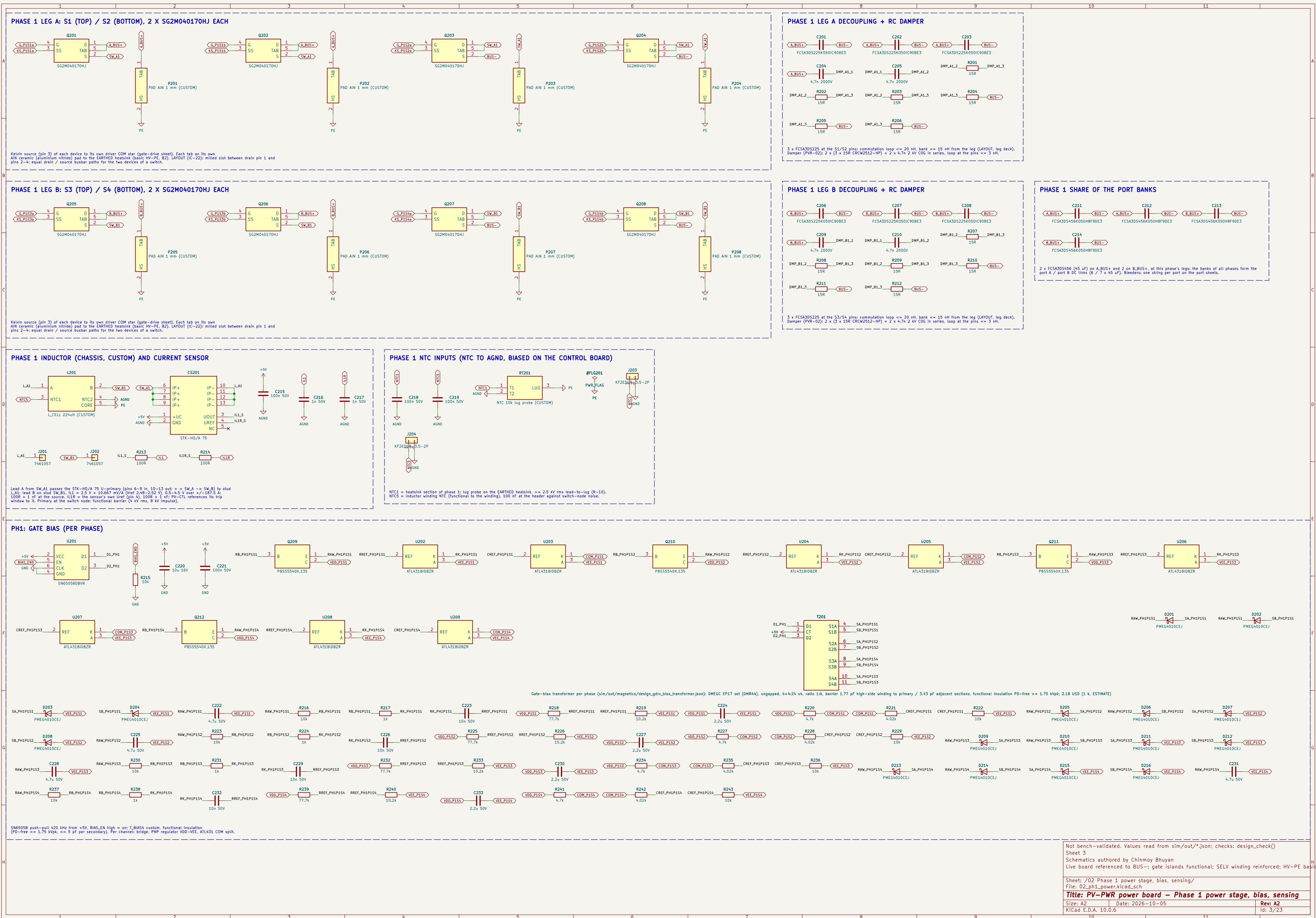


HOLD = A_HOLD OR B_HOLD ([Lport] >= 0.97-1.03 kA, contactor held), MOV_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below -200 V on its port). Port OC ([1] >= 387-413 A, lean_shunt) pulls FLT_N low (open drain); the control board's latch trips without firmware (4R4-fuse gap, port-4prc).
K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

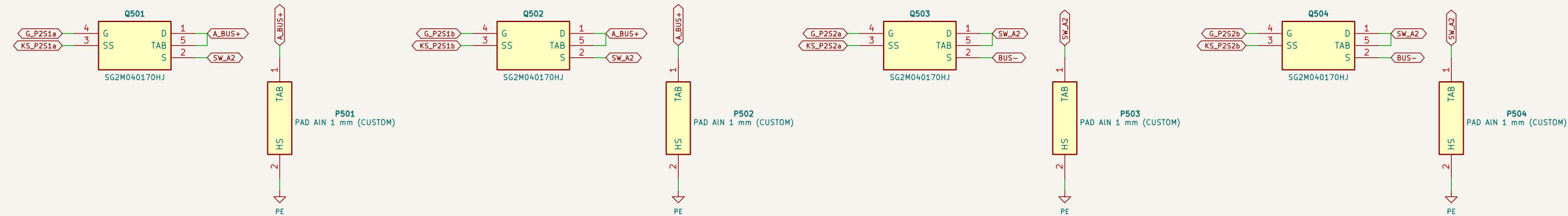
GROUND STAR POINTS



AGND meets GND at one point; GND meets BUS- at one point next to the port shunts (controller ground = BUS-); Layout: gate-drive primaries decoupled to GND locally.

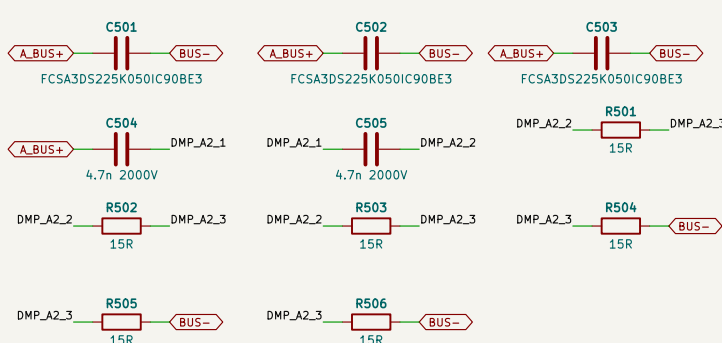


PHASE 2 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



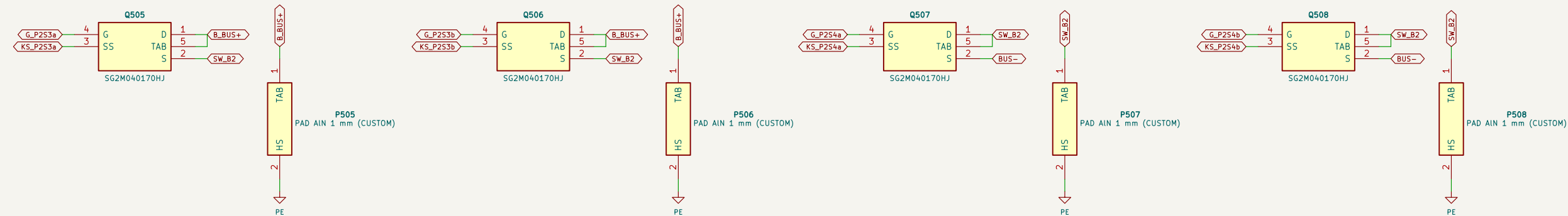
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG A DECOUPLING + RC DAMPER



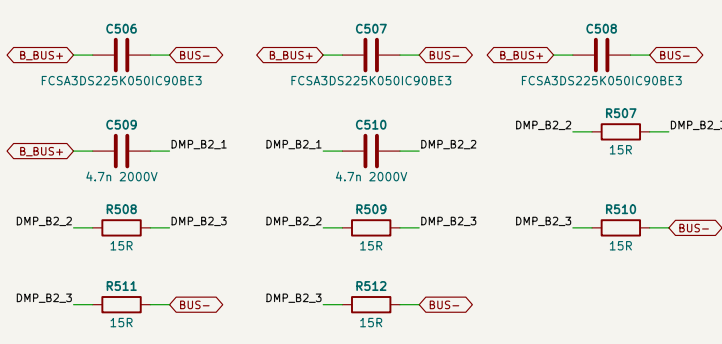
3 x FCSA3DS225 at the S1/S2 pins: commutation loop ≤ 20 nH, bank ≤ 15 nH from the leg (LAYOUT, leg deck)
 Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins ≤ 3 nH.

PHASE 2 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



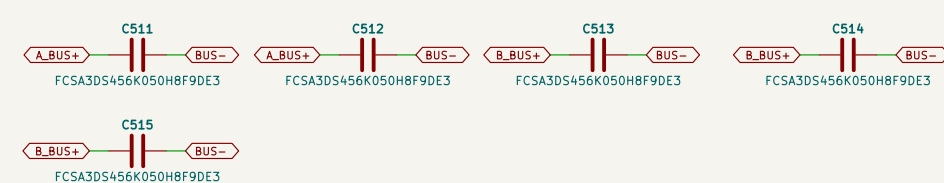
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG B DECOUPLING + RC DAMPER



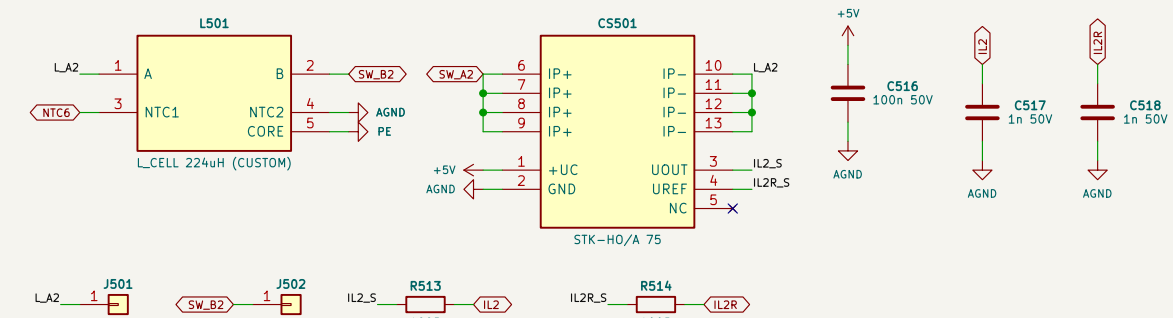
3 x FCSA3DS225 at the S3/S4 pins: commutation loop ≤ 20 nH, bank ≤ 15 nH from the leg (LAYOUT, leg deck)
 Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins ≤ 3 nH.

PHASE 2 SHARE OF THE PORT BANKS



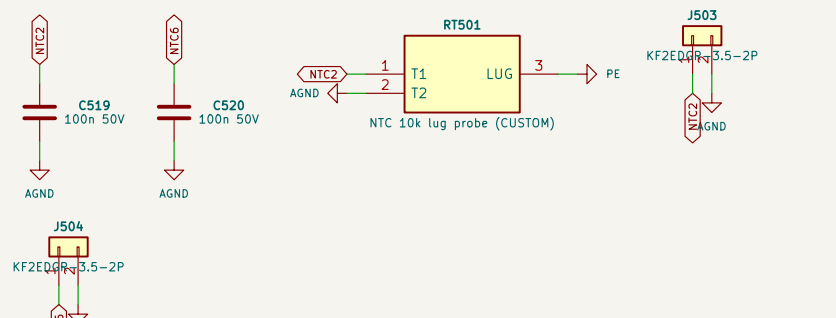
2 x FCSA3DS456 (45 uF) on A_BUS+ and 3 on B_BUS+, at this phase's legs: the banks of all phases form the port A / port B DC links ($6 / 7 \times 45$ uF); the third on B_BUS+ is the battery side's seventh (D-056: full-power load rejection). Bleeders: one string per port on the port sheets

PHASE 2 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



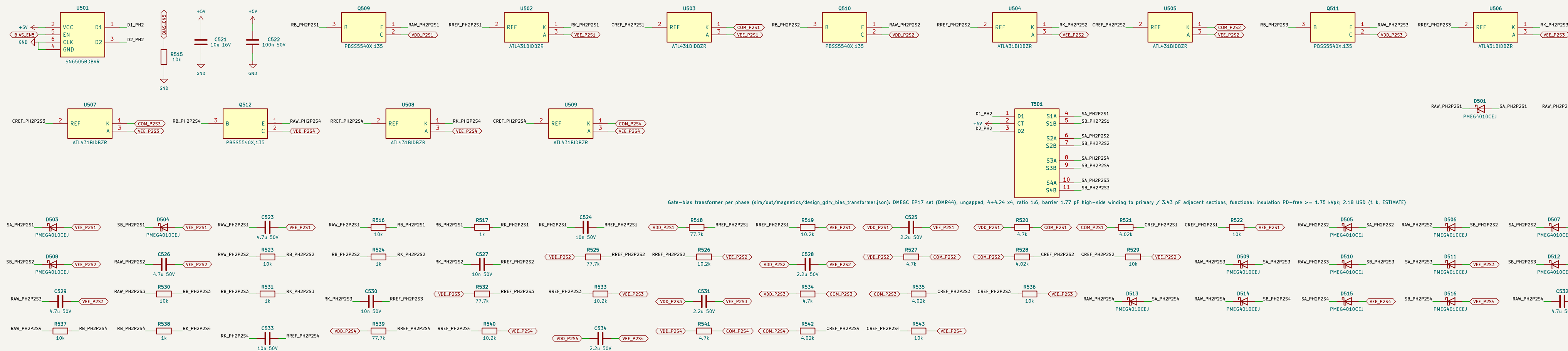
Lead A from SW_A2 passes the STK-H0/A 75 U-primary (pins 6-9 in, 10-13 out: + = SW_A -> SW_B) to stud L_A2; lead B on stud SW_B2. IL2 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A; 100K + 1 nF at the source. IL2R = the sensor's own Uref (pin 4). 100K + 1 nF: PV-CTL references its trip window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 2 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC2 = heatsink section of phase 2: lug probe on the EARTHED heatsink, ≥ 2.5 kV rms lead-to-lug (R-10)
NTC6 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

PH2: GATE BIAS (PER PHASE)



SN6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T_BIAS4 custom, functional insulation (PD-free ≥ 1.75 kVpk, ≤ 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 6
Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /05 Phase 2 power stage, bias, sensing/

File: 05_ph2_power.kicad_sch
Title: PV-PWR power board – Phase 2 power stage, bias, sensing

Size: A2	Date: 2026-10-05	Rev: A2
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KiCad E.D.A. 10.0.6 | Id: 6/23

P2S1: LOGIC SIDE

The schematic diagram illustrates the logic side of the P2S1 component. It features a central IC, U601 (SN74LVC1G170BVR), which is a 1-to-2 voltage level translator. The IC is connected to a +5V supply and ground. The input (A) is connected to DTS_P2S1, and the output (Y) is connected to DTB_P2S1. The IC is also connected to a network of resistors (R601, R602, R603, R604), capacitors (C601, C602, C603, C604, C605, C606), and logic inverters (D601, D602). The circuit includes power supply rails for +5V and GND, and various signal lines like INM_P2S1, DTS_P2S1, DTB_P2S1, and PWR_FLAG. A legend at the bottom explains the notation for complementary switches and timing parameters.

Legend:

- IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
- (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P251: DRIVER AND 2 GATES

The schematic diagram illustrates the internal circuitry of the P251 driver and its connection to various gates. The central component is the U602 driver chip (NSI651ASC-Q15WR), which is connected to the P251 pin headers. The driver's inputs are connected to the P251 pin headers, and its outputs are connected to the gates' inputs. The gates' outputs are connected to the P251 pin headers. The driver's inputs are connected to the P251 pin headers. The driver's outputs are connected to the gates' inputs. The gates' outputs are connected to the P251 pin headers. The driver's inputs are connected to the P251 pin headers. The driver's outputs are connected to the gates' inputs. The gates' outputs are connected to the P251 pin headers.

Per device: Ron/Reff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller clamp: N-MOSFET clamp (PMV30ENE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

[illegible][illegible][illegible]

[illegible]

P252: DRIVER AND 2 GATES

U604

+5V 15 VCC1 5 VDD_P252 6
 INP_P252 10 VCC2 4 OUTL_P252 4
 INN_P252 11 IN+ 7 CLAMP_P252 7
 EN5 14 RST/EN 7
 FLT_N 13 FLT 2 DK_P252 2
 R640 12 RDY 2 DESAT 2
 P16_P252 16 TEST 3 COM_P252 3
 GND 9 GND1 8 VEE_P252 8
 VEE2 1 P1_P252 1

NSI6651ASC-Q1SWR

Q609 PMV30ENAR
 CG_P252 1 3
 VELL_P252 1 3
 G_P252a 1 3
 SB_P252 1 3
 BG_P252 1 3
 VEE_P252 1 3

Q610 PMV30ENAR
 CG_P252 1 3
 VELL_P252 1 3
 G_P252a 1 3
 SB_P252 1 3
 BG_P252 1 3
 VEE_P252 1 3

Q609 PMV30ENAR
 CG_P252 1 3
 VELL_P252 1 3
 G_P252a 1 3
 SB_P252 1 3
 BG_P252 1 3
 VEE_P252 1 3

Q610 PMV30ENAR
 CG_P252 1 3
 VELL_P252 1 3
 G_P252a 1 3
 SB_P252 1 3
 BG_P252 1 3
 VEE_P252 1 3

Legend:

Per device: Ron/Off. 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
 Miller hold: N-MOSFET clamp (PMV30ENAR) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: to fet + 1u at the device, loop <= 1 nh.

[illegible]

P2S2: SHORT-CIRCUIT BOOSTER

DESAT > 10.3...10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V per device AO3400A + 22R pull the gate to VEE: BG holds -3 V (1k + FET gates).

[illegible]

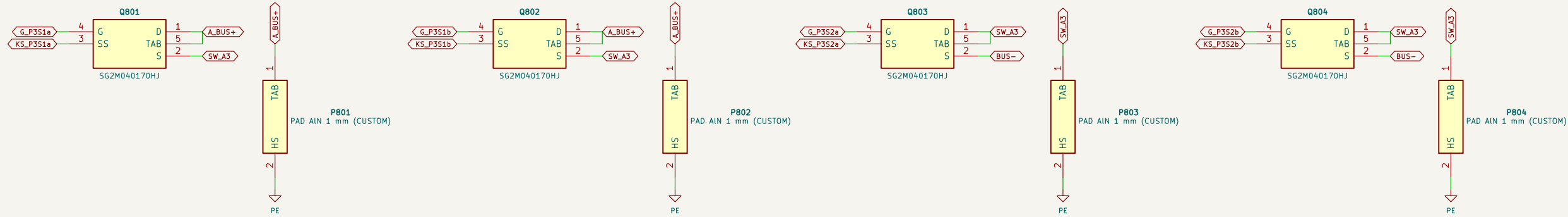
LEG A: VOLTAGE RULES

Sheet: /06 Phase 2 leg A gate drive (S1, S2)/
File: 06_ph2_legA_kick_06b

Size: A2	Date: 2026-10-05	Rev: A2
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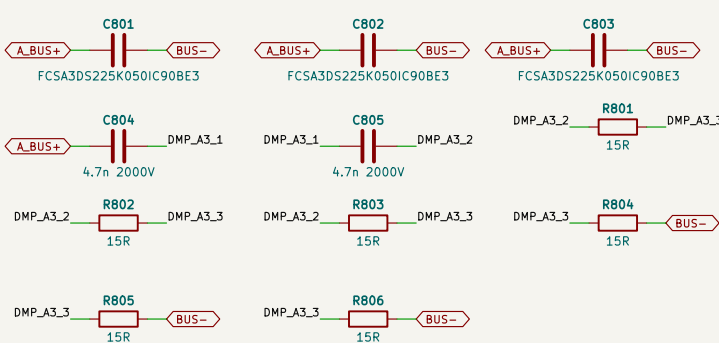
KiCad E.D.A. 10.0.6		Id: 7/23
10	11	

PHASE 3 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



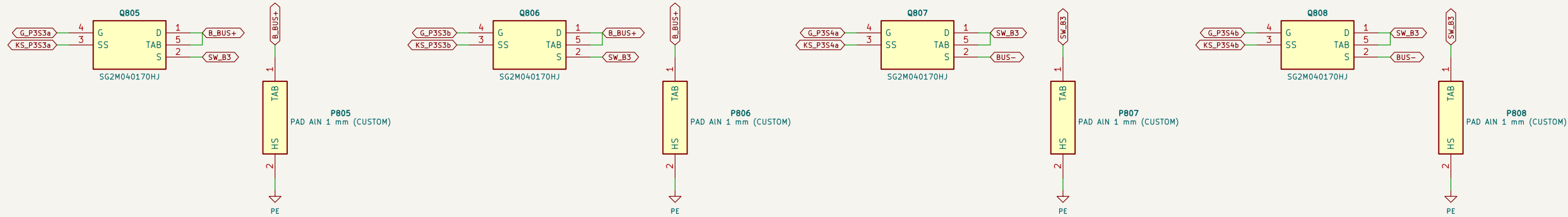
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG A DECOUPLING + RC DAMPER



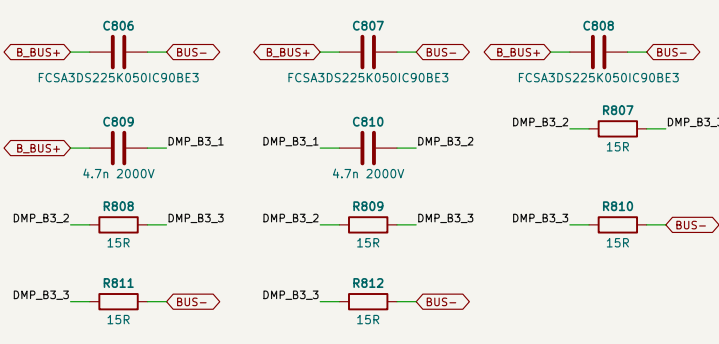
3 x FCSA3DS225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



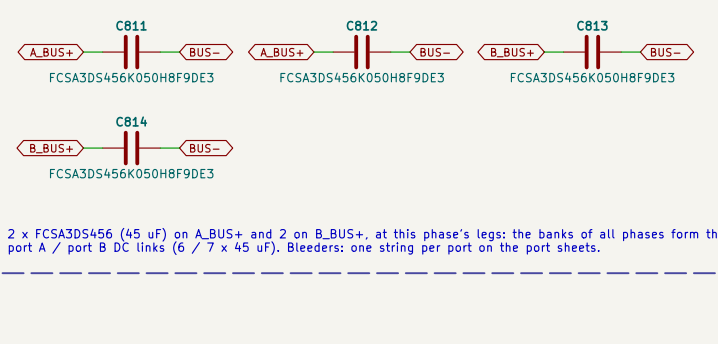
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG B DECOUPLING + RC DAMPER



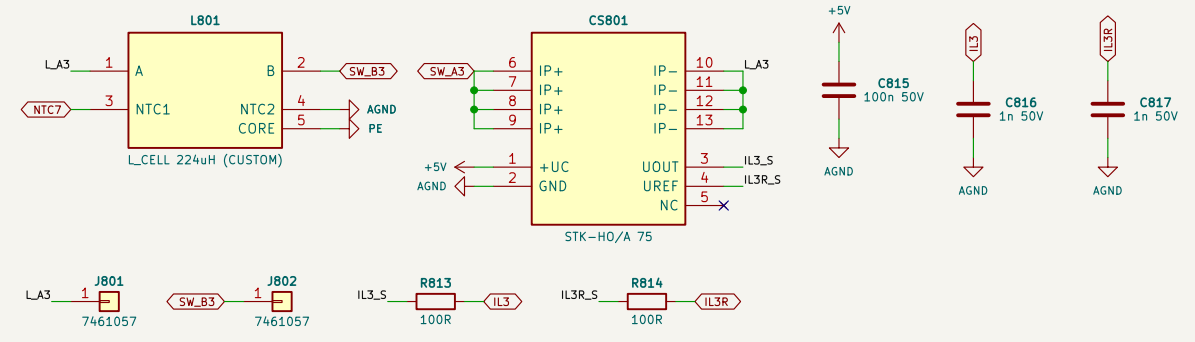
3 x FCSA3DS225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 3 SHARE OF THE PORT BANKS



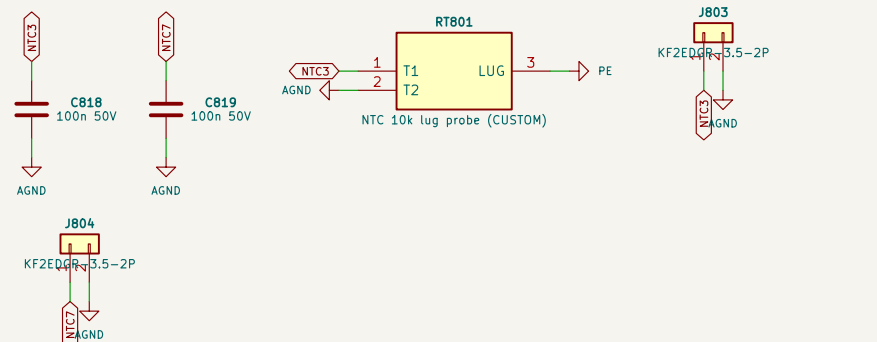
2 x FCSA3DS456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links (6 / 7 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 3 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



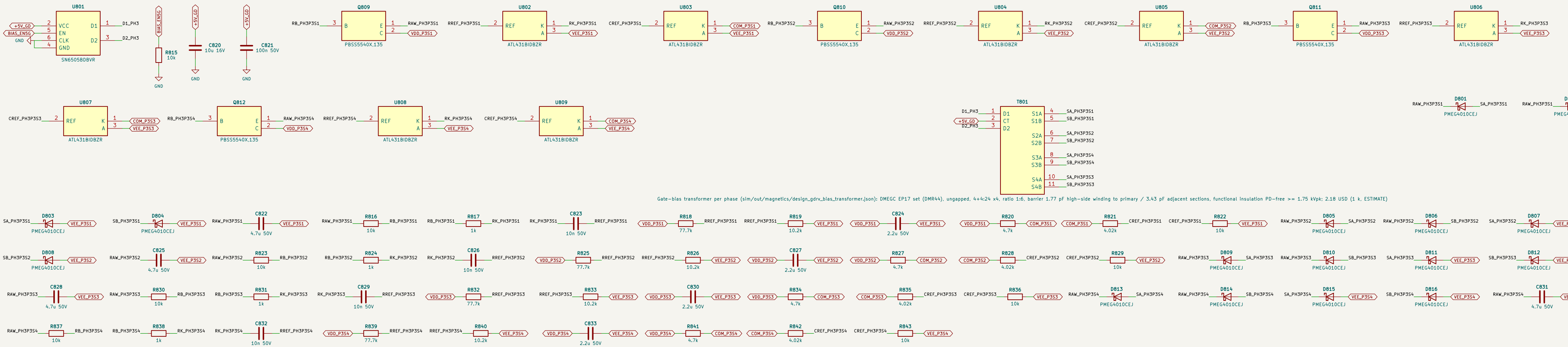
Lead A from SW.A3 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out; += SW.A -> SW.B) to stud
LA3: lead B on stud SW.B3, IL3 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V), 0.5-0.5 V over +/-187.5 A;
100R = 1 nF at the source, IL3R = the sensor's own tref (pin 6), 100R = 1 nF (PVC-CTL references its trip
window to IL. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 3 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



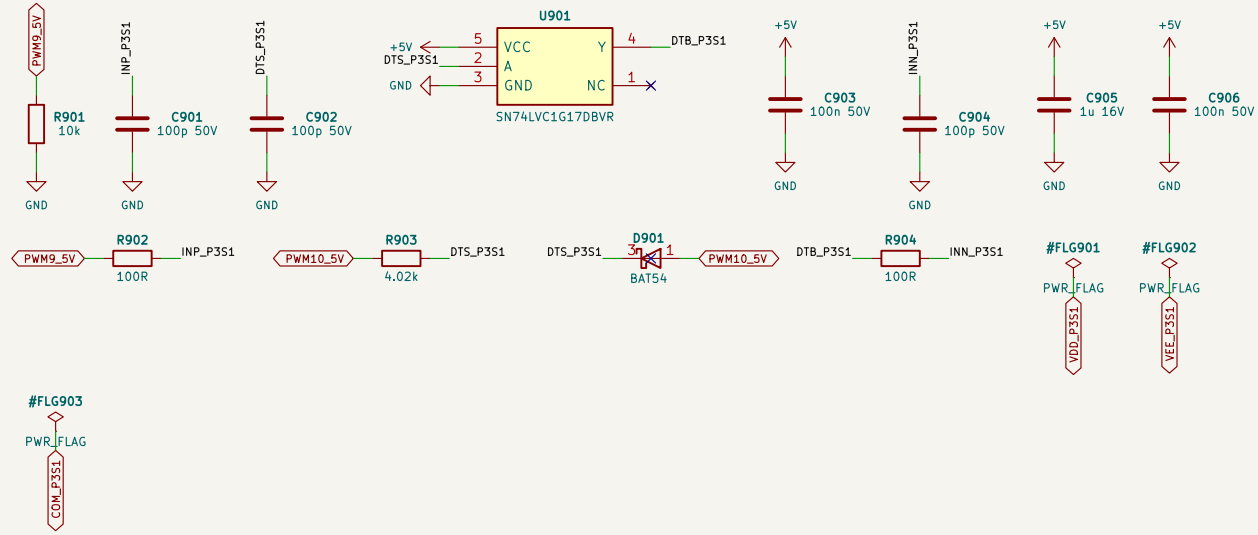
NTC3 = heatsink section of phase 3: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC7 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

PH3: GATE BIAS (PER PHASE)



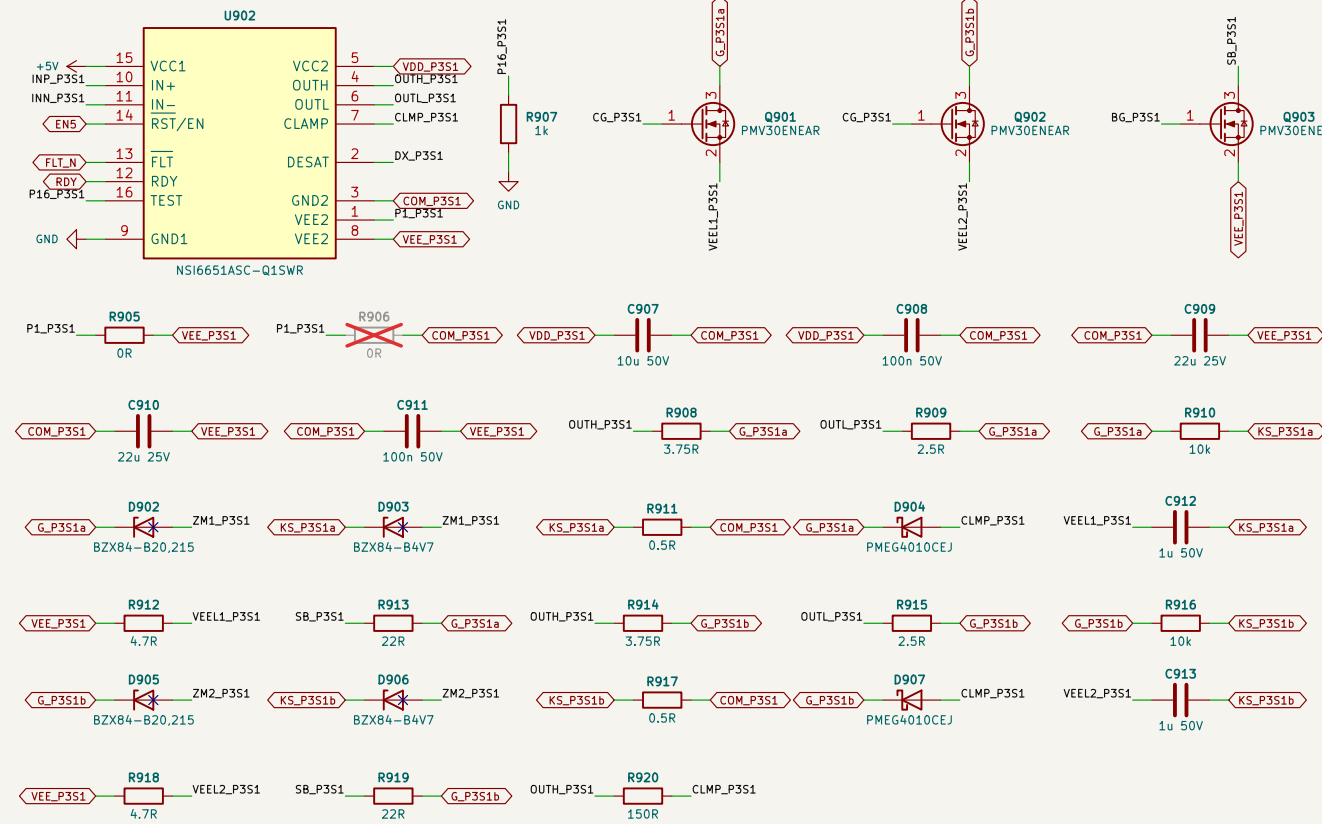
SN6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T.BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P3S1: LOGIC SIDE



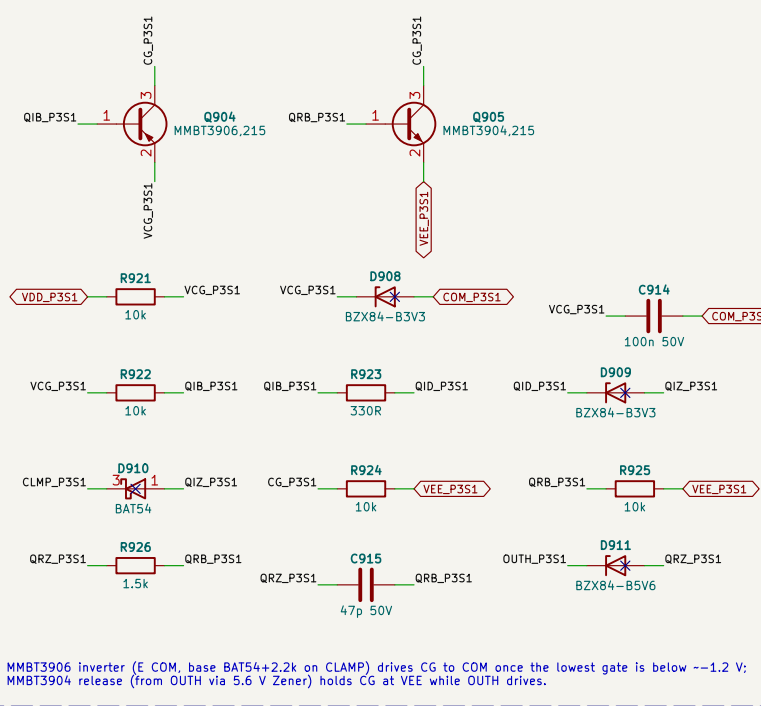
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S1: DRIVER AND 2 GATES



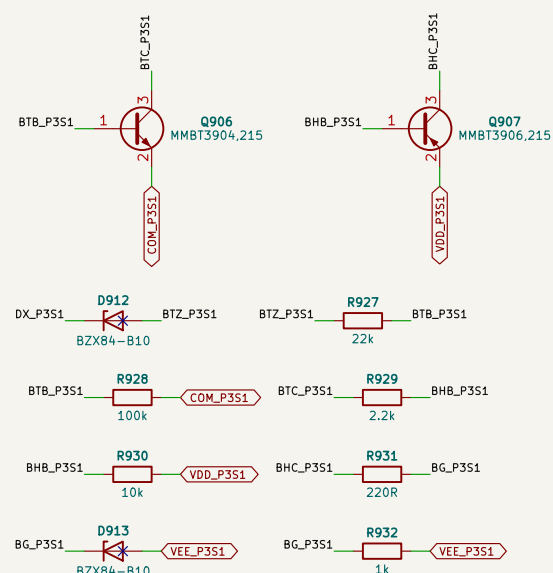
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEAE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S1: MILLER CLAMP CONTROL



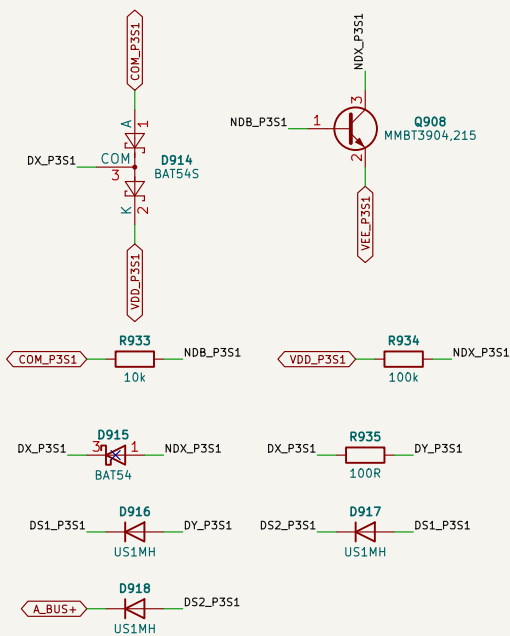
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S1: SHORT-CIRCUIT BOOSTER



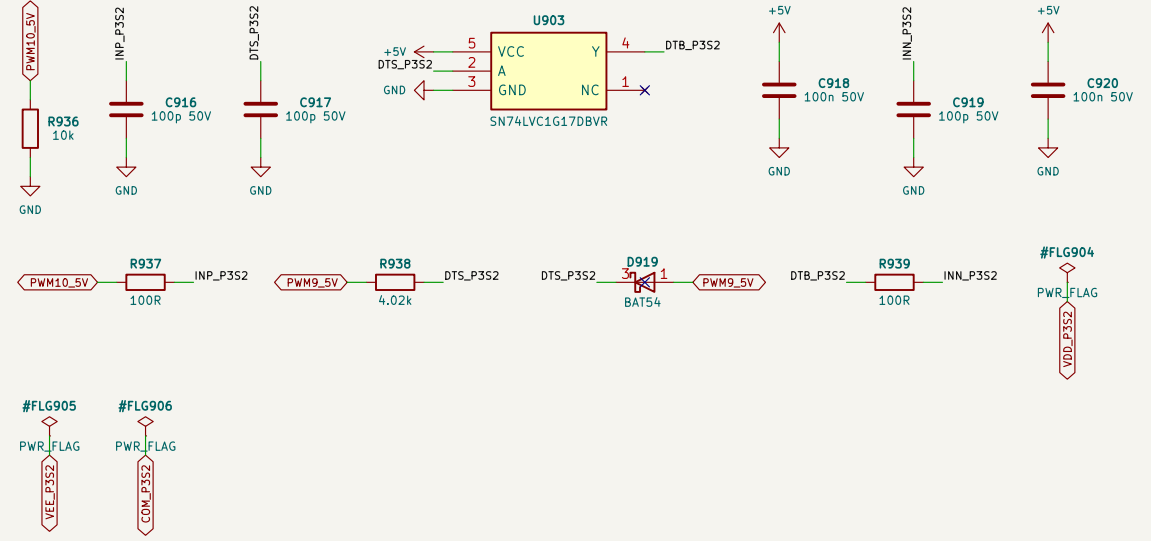
DESAT > 10.3--10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S1: DESAT



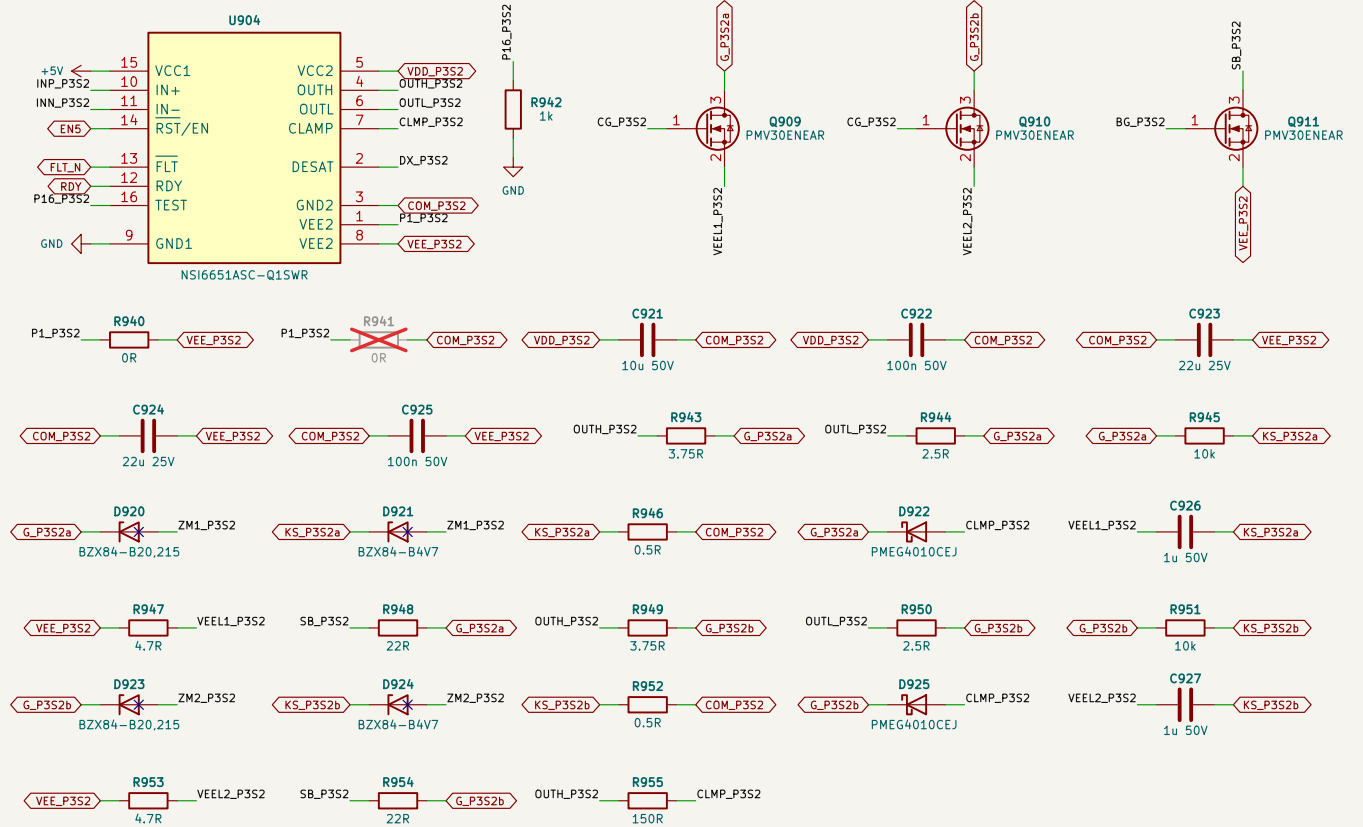
DESAT pin (internal 500 uA blanking source, trip 8.5--9.8 V) - Rs - 3 x US1MH - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

P3S2: LOGIC SIDE



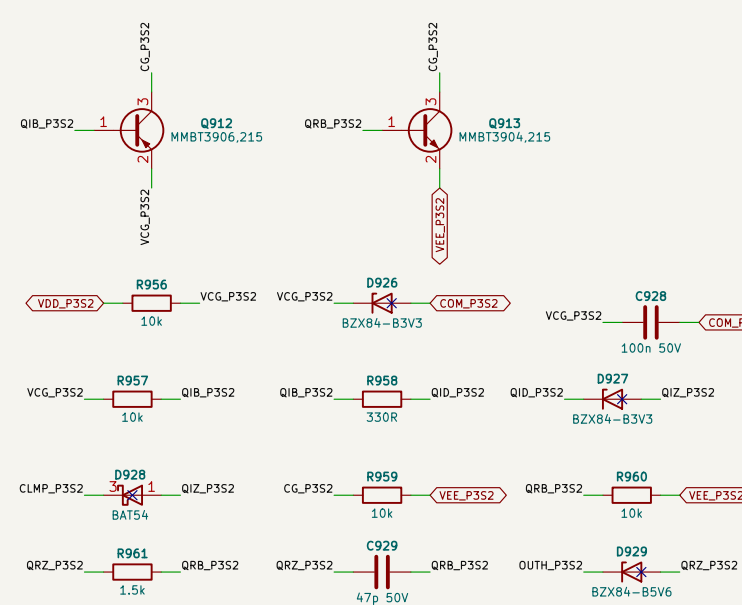
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S2: DRIVER AND 2 GATES



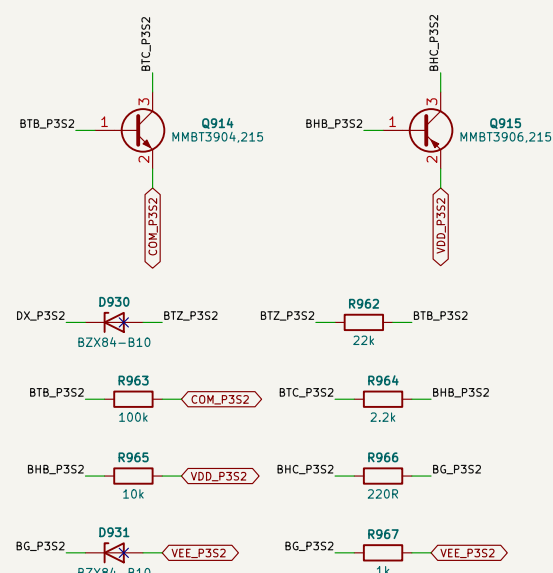
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEAE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S2: MILLER CLAMP CONTROL



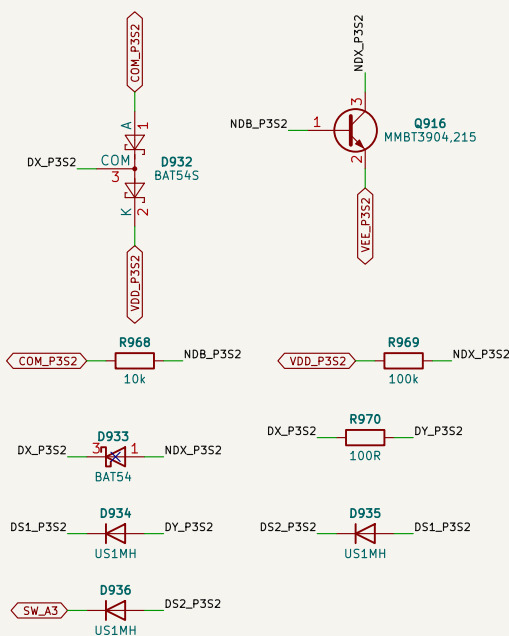
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3--10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S2: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5--9.8 V) - Rs - 3 x US1MH - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

LEG A: LAYOUT RULES

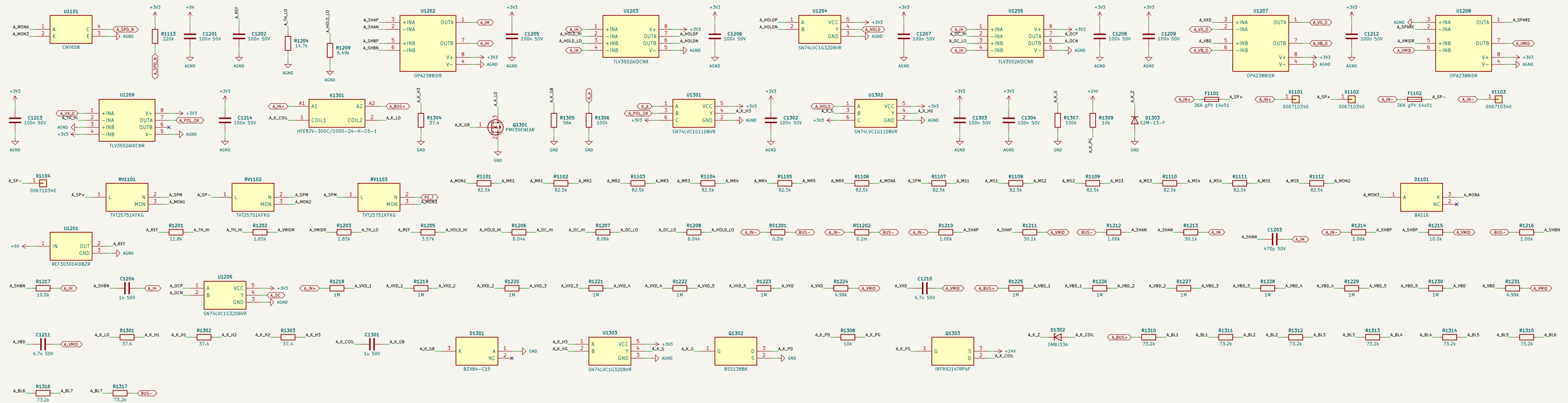
Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C.VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (opposite, zero): die +1.41 V vs VGS(n) min 1.94 V at 170 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{JN}- / A_{JN}-, bank A_{BUS}-. No fuse, no precharge; array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec_lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A_{JIN+} / A_{JIN-}, rank A_{BUS+}. No fuse, no precharge: array I_{sc} <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TVT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE.T. Up,eff 3793 V pole-PE and pole-pole at In 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D=032); I_{max} 25 kA. Branch fuses ETI CH14x51 pV36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 12
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE ba

Sheet: /11 Port A power path (pv)/
File: 11_leanA_power.kicad_sch

Title: PV-PWR power board – Port A power path (pv)

Size: A2	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 12/23

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 13
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /12 Port A sensing and interlocks/
File: 12_leanA_ctrl.kicad_sch

Title: **PV-PWR power board – Port A sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 13/23

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 14
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /13 Port A coil drives/
File: 13_leanA_drive.kicad_sch

Title: **PV-PWR power board – Port A coil drives**

Size: A3Date: 2026-10-05

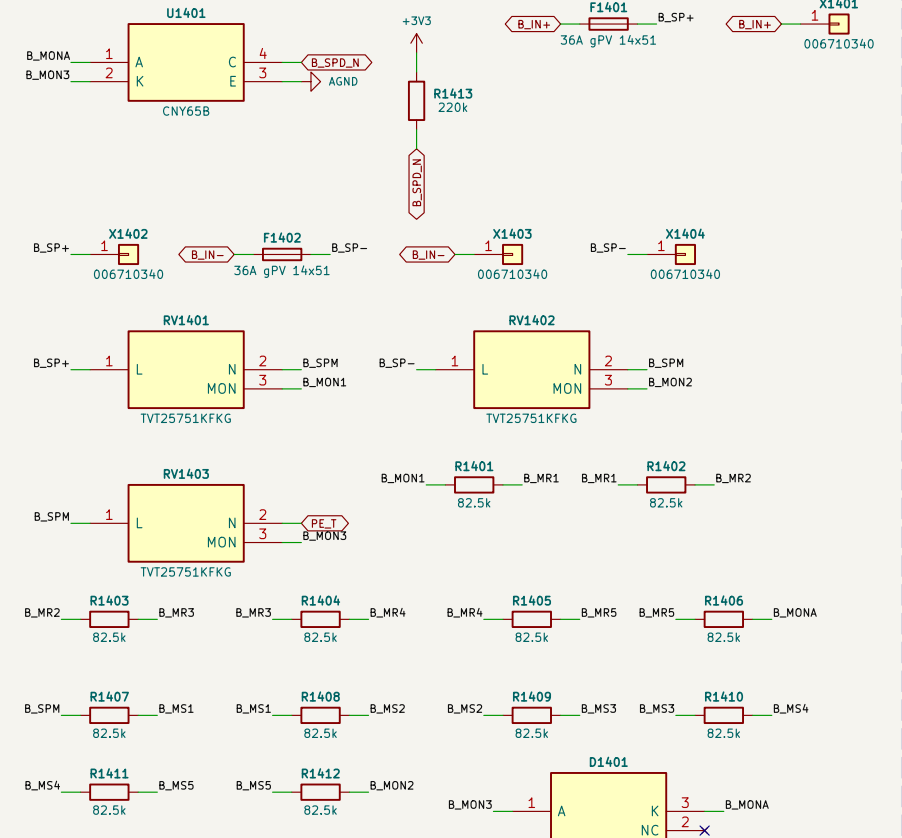
KiCad E.D.A. 10.0.6

Rev: **A2**
Id: 14/23

PORT B (BATTERY SOURCE), D-044 LEAN PORT

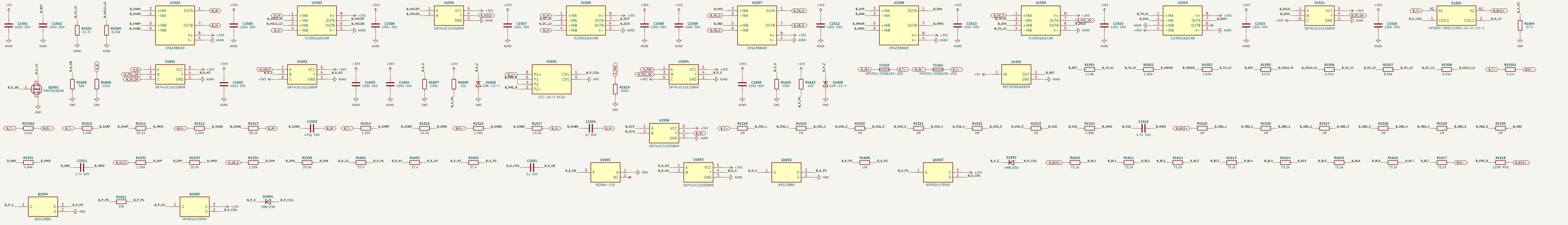
Terminal side R_MN+ / R_MN- bank R_BUS+/- HPE501 250 A at in both poles (breaks 1.25-50 A at in).
Contactor opens <= 0.97 A, held above (port_Lapc time). Installation: upstream circuiters 987-1250 A within 0.26 s.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TV25751 = 3 (605 VDC, thermal time = monitor load); MOV1 + PM MOV2 - TV MOV3 W/PEL1 Use at 3795 V pole-PE and post-pole at 3 kA (GMC, <= 4.0 VV => reinforced 5 VV / basic 4 V, D-032); max 25 mA Branch
From ESI CHAN55: 8P, 50 A, 50 A (D/F 2 ms); Monitor loop through 500 Tera and all time (data) -> CHAN55
-> R81 (sum on the ID line); R81 = 1 at rest above 200 V = varistor network open -> CTR1 inhibit/stop.

AR FUSES (BATTERY-TYPE PORT)



HPE501/0008100-250 in both poles breaks 1.25-50 A at only (partial range);
the part QC trip and the contactor keep it out of 250 A-1.25 A.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 16
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /14 Port B sensing and interlocks/
File: 14_leanB_ctrl.kicad_sch

Title: **PV-PWR power board – Port B sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 16/23

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 17
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /15 Port B coil drives/
File: 15_leanB_drive.kicad_sch

Title: **PV-PWR power board – Port B coil drives**

Size: A3

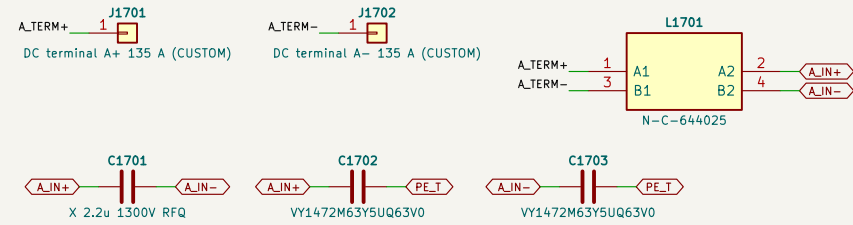
Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

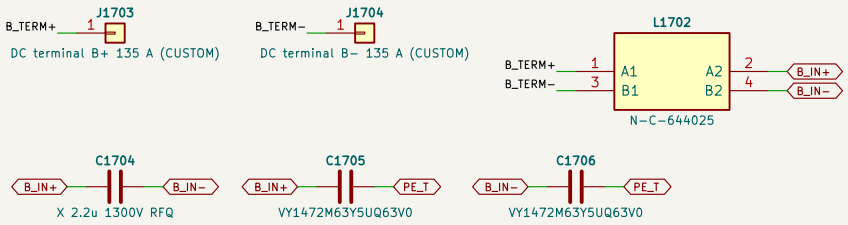
Id: 17/23

PORT A TERMINALS AND EMI FILTER



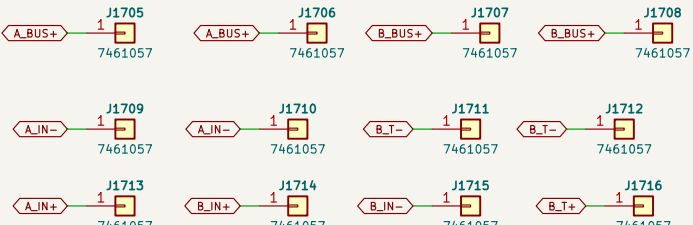
PV: chassis terminals A_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps A_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



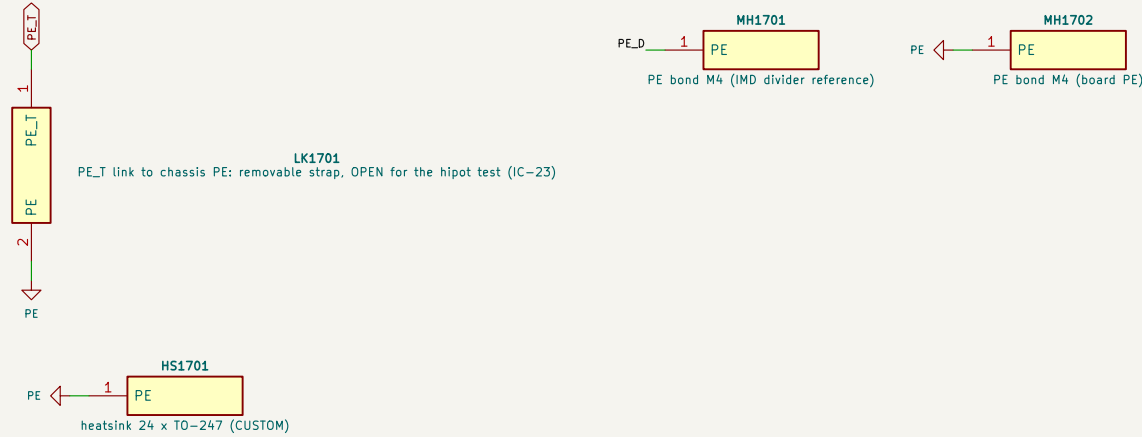
battery: chassis terminals B_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- / B_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



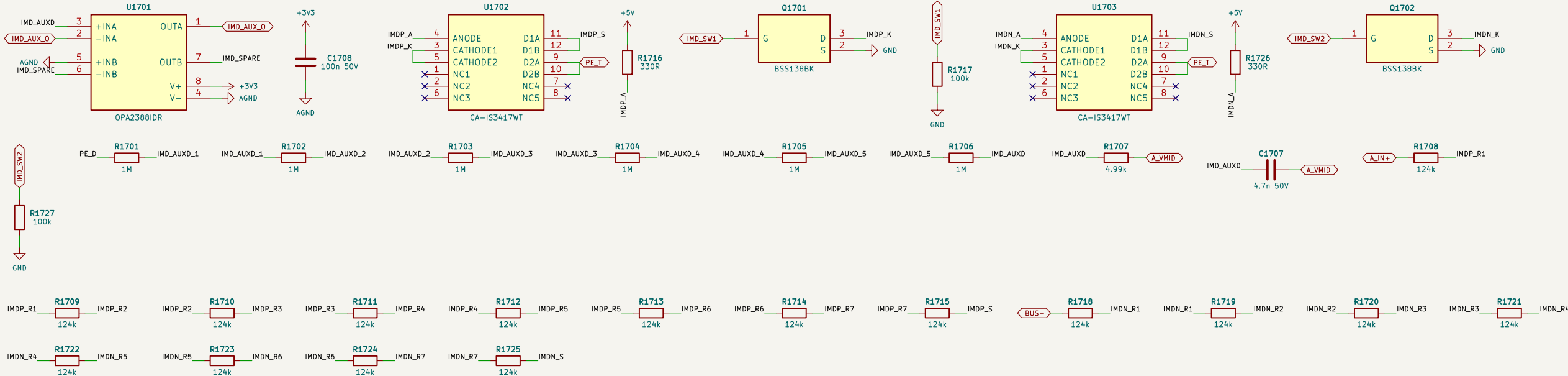
Bus entries from the contactors / fuses: A_BUS+, B_BUS+, A_IN- and B_T- (into the shunts), two studs each; taps A_IN+, B_IN+, B_IN-, B_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE_D = IMD divider bond; the heatsink is earthed at two points.

INSULATION MONITOR (PORT.LEAN_IMD)



IMD_SW1 = string A+ -> PE, IMD_SW2 = string PE -> BUS- (high = connected);
VPE = PE divider against BUS- (VMID of port A).

Not bench-validated. Values read from sim/output/*.json; checks: design_check()

Sheet 18

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /15 Terminals, EMI filter, insulation test/

File: 15_terminals_imd.kicad_sch

Title: PV-PWR power board - Terminals, EMI filter, insulation test

Size: A3

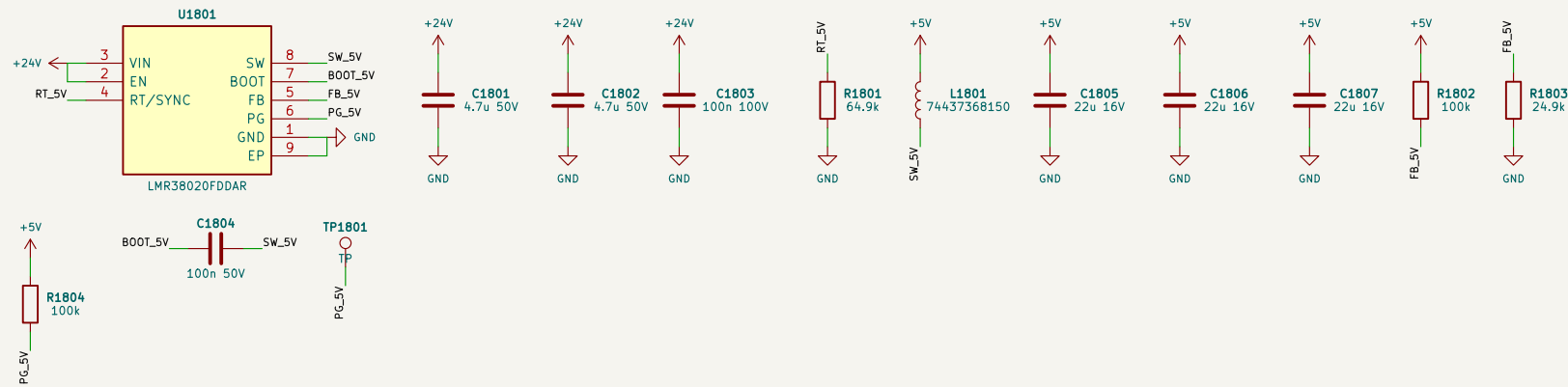
Date: 2026-10-05

Rev: A2

KiCad E.D.A. 10.0.6

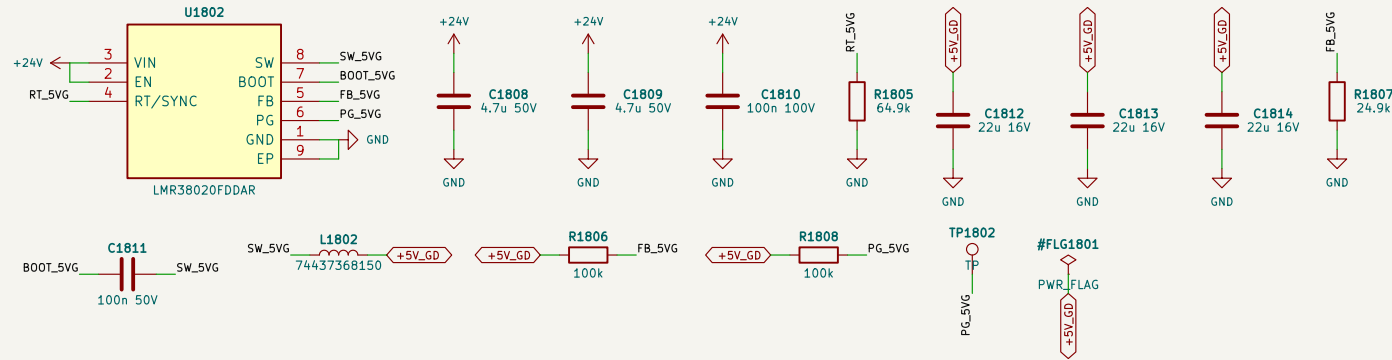
Id: 18/23

24 V -> +5V (LMR38020, 400 KHZ, 2 A)



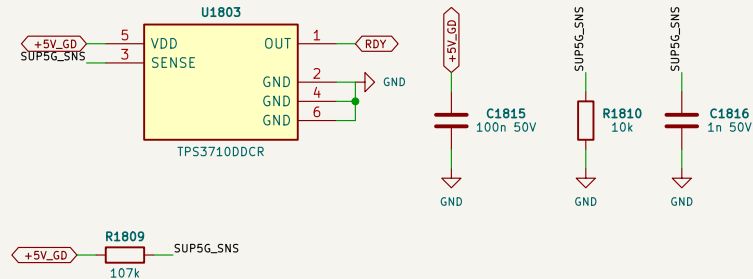
NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2, ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)



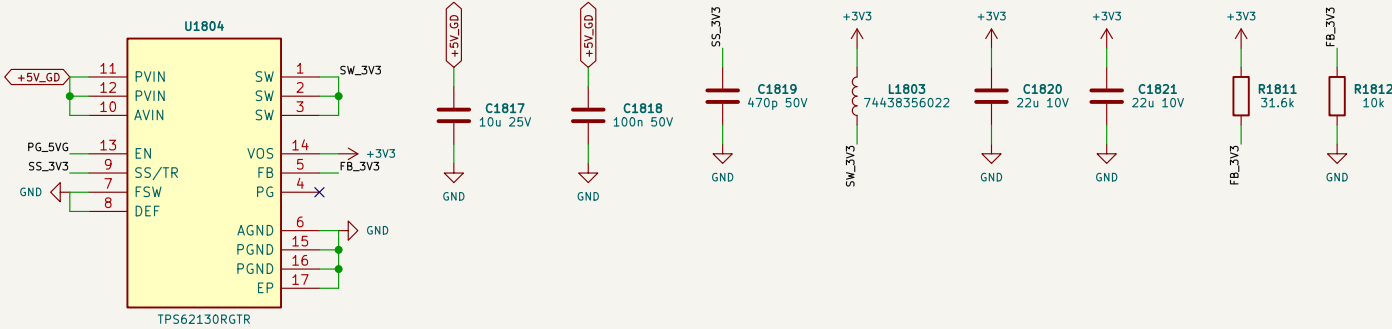
gate bias of phases 3 only, ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD SUPERVISOR ON RDY



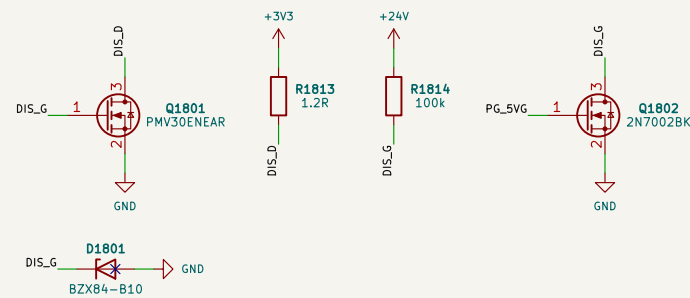
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3 is below 4.52-4.69 V.

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP



EN = PG_SVG, FB 31.6k / 10k 0.1 % -> 3.240-3.433 V (>= 3.20 V for the control board's reference), SS/TR 470pF COG; ramp-up 12-16 mV/us (F28003x VDDIO 8-100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control <= 157 mA).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG_SVG low (+5V_GD < 90-94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R: +3V3 falls at >= 20 mV/us through the BOR region (F28003x SRVDDIO-DN).

SELV 24 V TO THE CONTROL BOARD (J2)



AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 20.3-29.8 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12-36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()

Sheet 19

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /16 Live 5 V / 3.3 V, SELV connector/

File: 16_supplies.kicad_sch

Title: PV-PWR power board - Live 5 V / 3.3 V, SELV connector

Size: A3

Date: 2026-10-05

Rev: A2

KiCad E.D.A. 10.0.6

Id: 19/23

TAP A_IN+



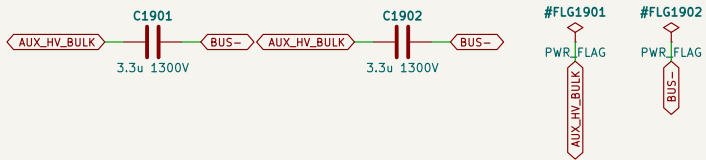
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

TAP B_T+



2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual -> 1199 V (UR 1300 V); filter corner 546 Hz.

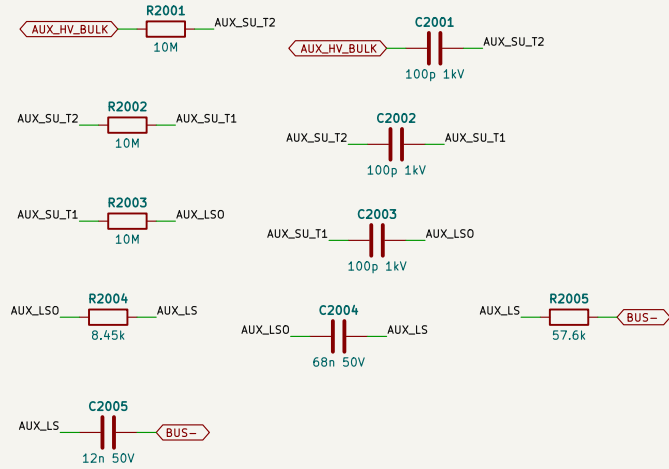
Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 20
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /17 AUX75 input taps, OR, film/
File: 17_aux75_input.kicad_sch

Title: PV-PWR power board – AUX75 input taps, OR, film

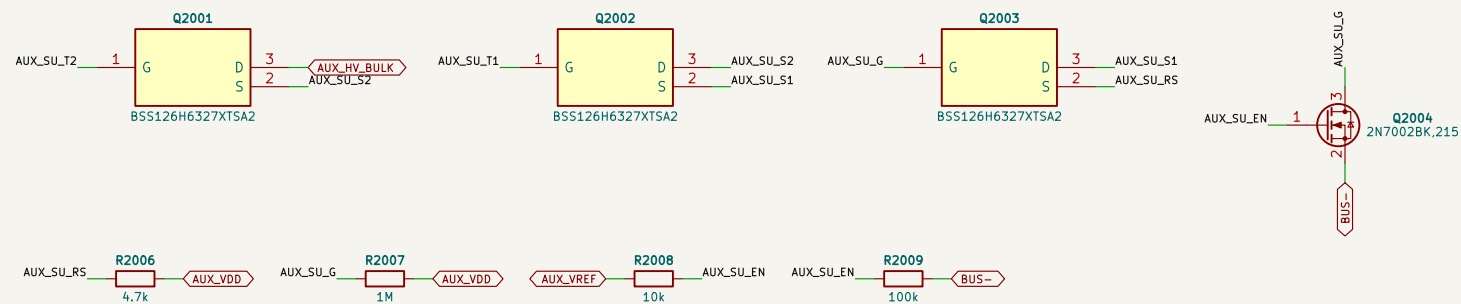
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 20/23

COMPENSATED TAP STRING



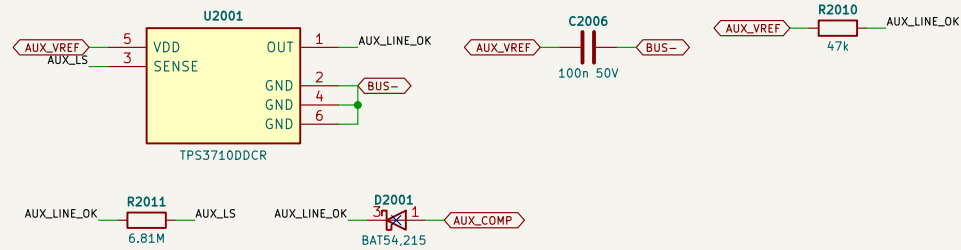
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



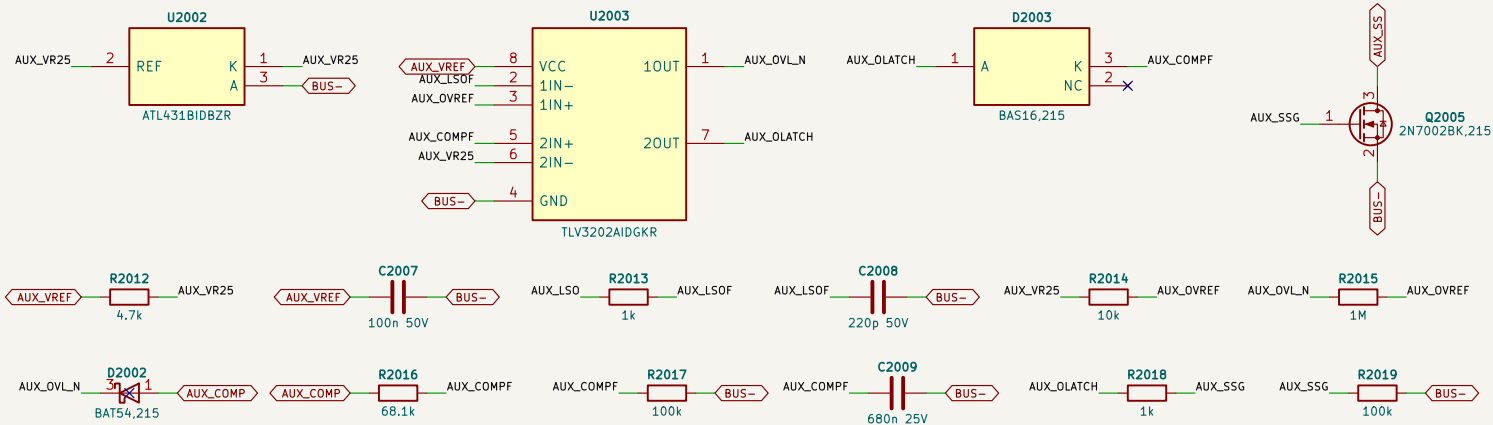
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



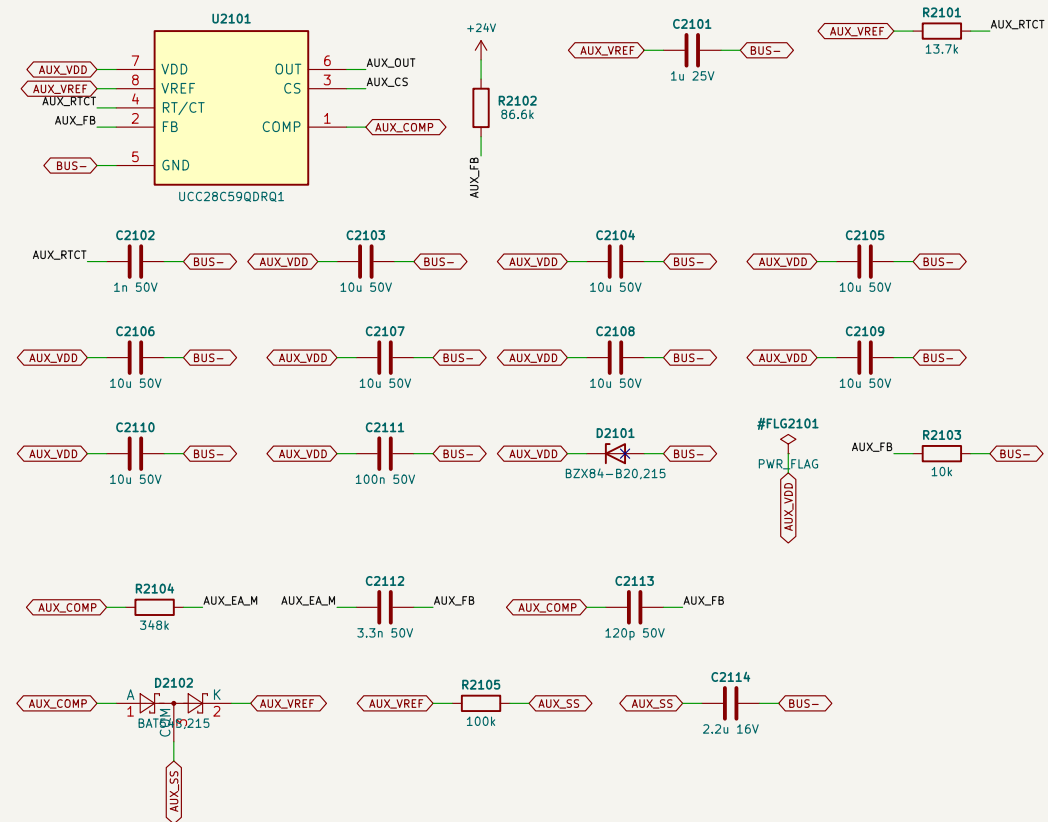
LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

INPUT OV LOCKOUT AND OVERLOAD TIMER



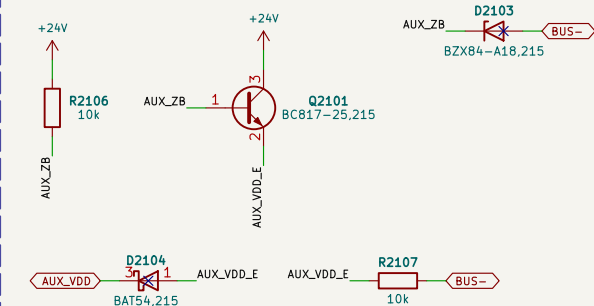
Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 103 / 160 / 256 W (no trip at 75 W + 6 W pull-in); timer 35–80 ms -> latch -> SS crowbar -> hiccup.

PWM CONTROLLER AND ERROR AMPLIFIER



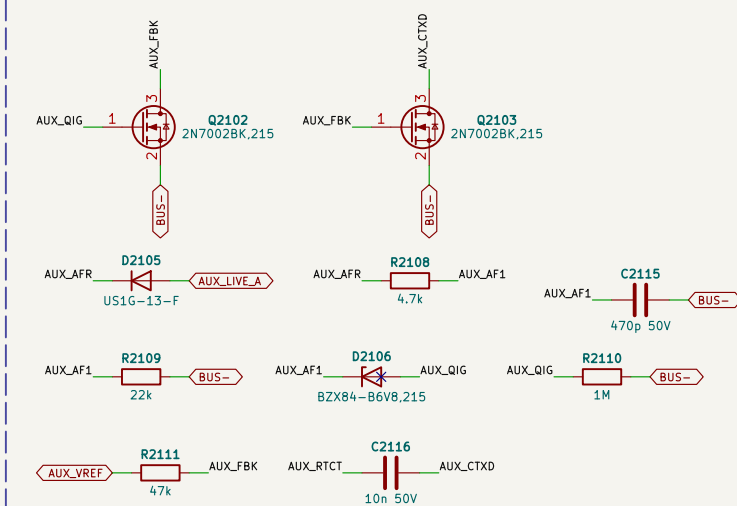
live 24 V into FB: 23.62 / 24.15 / 24.68 V (VFB 2.45–2.55 V, 0.1 %)
type II: mid-band gain 4.0; crossover 241–667 Hz, phase margin ≥ 58 deg type II zero 139 Hz type II pole 3811 Hz

VDD FOLLOWER FROM THE LIVE 24 V



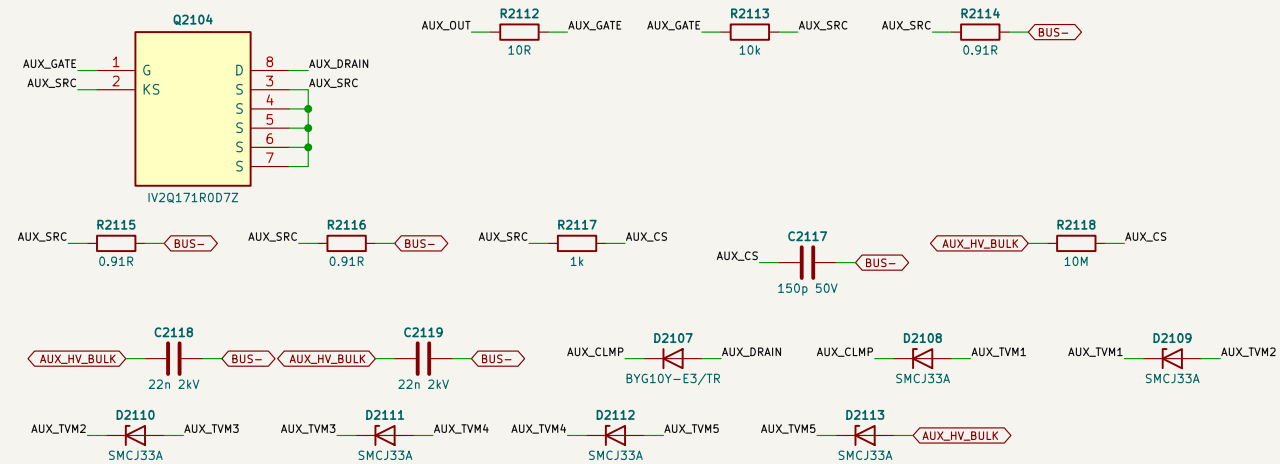
(30 W case, unchanged) follower Zener bias: VDD = gate drive 15.46–18.11 V (BZX84–A18, –40..85 °C)

FREQUENCY FOLDBACK ON THE LIVE WINDING PLATEAU (AX-03)



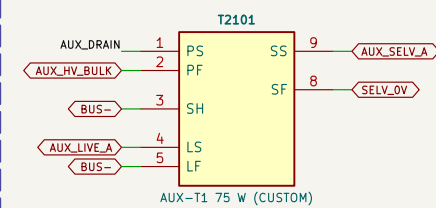
(30 W case, unchanged) foldback: switched onto RT/CT below 7.7–10.1 V output → fsw 5.9 kHz (5.3–6.8)

SIC SWITCH, CURRENT SENSE, LINE FEED-FORWARD, CLAMP



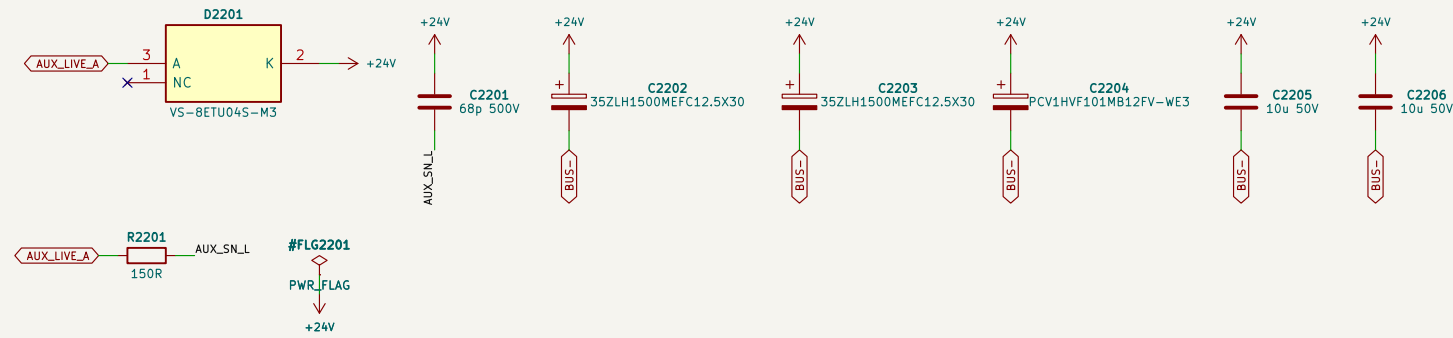
3 In parallel = 0.303333R: lowest limit 2.90 A delivers 103 W ($>= 110\%$ of 75 W) at every input; highest 3.91 A with the line feed-forward (4.27 A without) line feed-forward HV_BULK -> CS (CRH2512): CS offset 101 mV at 1000 V cancels most of the 155-250 nS current delivery overshoot, so the highest current limit no longer rises with the input Clamp: BYG10Y + 6 x SMCJ33A; V_DS 1316 / 1420 V at 1000 / 1100 V.

AUX-T1 (CUSTOM – THE MODULE'S REINFORCED SUPPLY BARRIER B1)



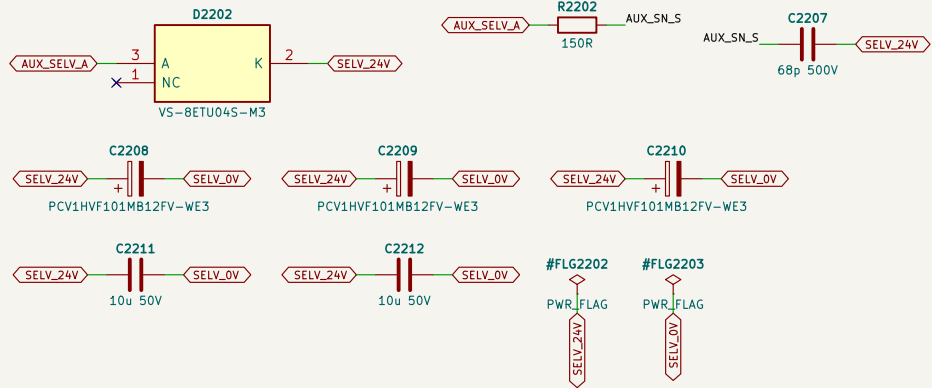
Lp 0.485 mH +/- 7 %; shield and live winding on BUS-; SELV winding reinforced (PD <= 10 pC routine test).
Electrical requirement: aux75_spec.json transformer_requirement.

LIVE 24 V RECTIFIER, HOLD-UP



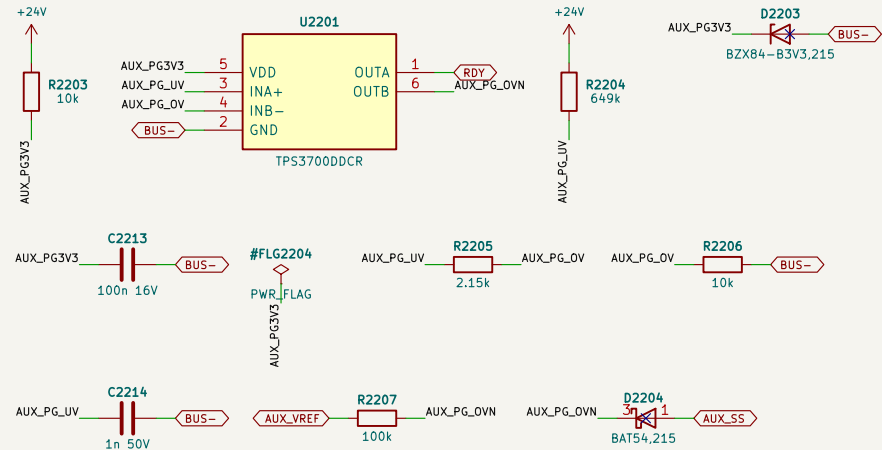
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02–21.81 V (status), OV rises 26.13–26.76 V (SS pulled low)